NATIONAL MATHEMATICS CONTEST

A-LEVEL PAPER 1.

SATURDAY 18TH MARCH, 2017 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) *Answer scripts of only qualified students should be returned to the contest coordinator, latest* **Monday 20th March, 2017**.

(d) **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 29*th July,* 2017 *at NOON at Makerere University, Kampala.*

(e) *National Mathematics Contest* 2017 *Paper* 2 *will be done on* 17*th June,* 2017 *at various centres.*

SECTION A : 5 marks each

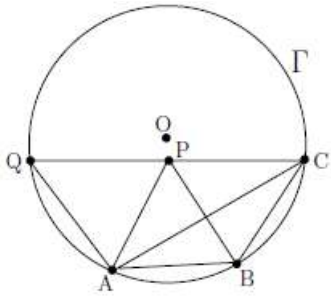
1. Determine the number a such that $\log_a(10) + \log_a\left(10^2\right) + \cdots + \log_a\left(10^{10}\right) = 110$.

2. Let (a, b, c) be a Pythagorean triple, i.e., a triplet of positive integers with $a^2 + b^2 = c^2$. Prove that $\left(\frac{c}{a} + \frac{c}{b}\right)^2 > 8$.

3. Solve the following system of equations. Carefully justify your answer.

$$\frac{4x^2}{1+4x^2} = y, \quad \frac{4y^2}{1+4y^2} = z, \quad \frac{4z^2}{1+4z^2} = x$$

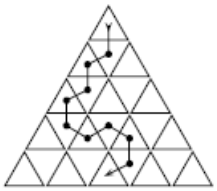
4. Let Γ be a circle with radius r. Let A and B be distinct points on Γ such that $AB < \sqrt{3}r$. Let another circle with centre B and radius AB meet Γ again at C. Let P be the point inside Γ such that triangle ABP is equilateral. Finally, let CP meet Γ again at Q. Prove that $PQ = r$.



5. Find all real solutions to the equation $4x^2 - 40[x] + 51 = 0$. Here, if x is a real number, then $[x]$ denotes the greatest integer that is less than or equal to x say $[3.7] = 3, [-2.1] = -3, [5] = 5$.

6. Evaluate the sum $\displaystyle\sum_{n=1}^{1994} (-1)^n \frac{n^2+n+1}{n!}$.

7. Consider an equilateral triangle of side length n, which is divided into unit triangles, as shown. Let $f(n)$ be the number of paths from the triangle in the top row to the middle triangle in the bottom row, such that adjacent triangles in our path share a common edge and the path never travels up (from a lower row to a higher row) or revisits a triangle. An example of one such path is illustrated below for $n = 5$. Determine the value of $f(2005)$.



8. Twenty five boys and twenty five girls sit around a table. Prove that, it is always possible to find a person both of whose neighbours are girls.

9. Let $ABCD$ be a convex quadrilateral with

$$\angle CBD = 2\angle ADB, \angle ABD = 2\angle CDB, \text{and} \quad AB = CB.$$

Prove that $AD = CD$.

10. If α, β, γ are roots of $x^3 - x - 1 = 0$, compute $\dfrac{1+\alpha}{1-\alpha} + \dfrac{1+\beta}{1-\beta} + \dfrac{1+\gamma}{1-\gamma}$.

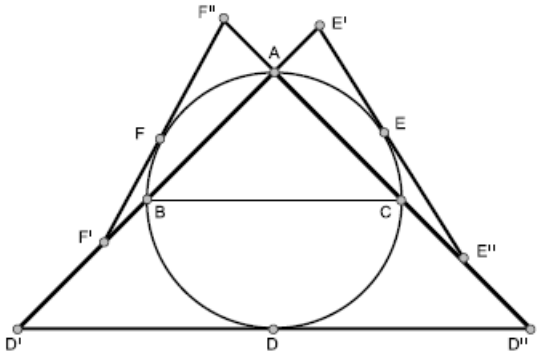
SECTION B : 10 **marks each**

11. How many ways can 8 mutually non-attacking rooks be placed on the 9×9 chessboard (shown here)



so that all 8 rooks are on squares of the same colour? [Two rooks are said to be attacking each other if they are placed in the same row or column of the board.]

12. The vertices of a right triangle ABC inscribed in a circle divide the circumference into three arcs. The right angle is at A, so that the opposite arc BC is a semicircle while arc AB and arc AC are supplementary. To each of the three arcs, we draw a tangent such that its point of tangency is the midpoint of that portion of the tangent intercepted by the extended lines AB and AC. More precisely, the point D on arc BC is the midpoint of the segment joining the points D' and D'' where the tangent at D intersects the extended lines AB and AC. Similarly for E on arc AC and F on arc AB. Prove that triangle DEF is equilateral.



13. Let $\mathbb{N}$ be a set of natural numbers $\mathbb{N} = \{0, 1, 2, \ldots\}$. Determine all functions $f : \mathbb{N} \to \mathbb{N}$ such that $xf(y) + yf(x) = (x + y)f\left(x^2 + y^2\right)$ for all x and y in $\mathbb{N}$.

14. Find all real positive solutions (if any) to

$$\begin{aligned} x^3 + y^3 + z^3 &= x + y + z \\ x^2 + y^2 + z^2 &= xyz \end{aligned}$$

15. (a) Show that at any party there are two people who have the same number of friends at the party (assume that all friendships are mutual).

    (b) Seven darts are thrown onto a circular dartboard of radius 10 units. Show that there will always be two darts which are at most 10 units apart?

SECTION A : 5 marks each

1. By the rules for logs, the expression given is

$$\begin{aligned}\log_a\left(10\cdot 10^2\cdot 10^3\cdots 10^{10}\right) &= \log_a\left(10^{1+2+3+\cdots+10}\right)\\ &= \log_a\left(10^{\frac{10\cdot 11}{2}}\right)\\ &= \log_a\left(10^{55}\right)\\ &= 55\log_a(10)\\ &= \frac{55}{\log_{10}(a)}\end{aligned}$$

Therefore, from

$$\frac{55}{\log_{10}(a)} = 110 \quad\Rightarrow\quad \log_{10}(a) = \frac{1}{2}$$

so that

$$a = \sqrt{10}$$

2. *Alternative Solution 1*

Let (a, b, c) be a Pythagorean triple. View a, b as lengths of the legs of a right angled triangle with hypotenuse of length c; let θ be the angle determined by the sides with lengths a and c. Then

$$\begin{aligned}\left(\frac{c}{a}+\frac{c}{b}\right)^2 &= \left(\frac{1}{\cos\theta}+\frac{1}{\sin\theta}\right)^2 \\ &= \frac{\sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta}{(\sin\theta\cos\theta)^2} \\ &= 4\left(\frac{1+\sin 2\theta}{\sin^2 2\theta}\right) \\ &= \frac{4}{\sin^2 2\theta}+\frac{4}{\sin 2\theta}\end{aligned}$$

Note that because $0 < \theta < 90°$, we have $0 < \sin^2\theta \leq 1$, with equality only if $\theta = 45°$. But then $a = b$ and we obtain $\sqrt{2} = c/a$, contradicting a, c both being integers. Thus, $0 < \sin 2\theta < 1$ which gives $\left(\frac{c}{a}+\frac{c}{b}\right)^2 > 8$.

Alternative Solution 2

Defining θ as in Solution 1 above, we have $\frac{c}{a}+\frac{c}{b} = \sec\theta+\csc\theta$. By the AM-GM inequality, we have $\frac{(\sec\theta+\csc\theta)}{2} \geq \sqrt{\sec\theta\csc\theta}$. So

$$\frac{c}{a}+\frac{c}{b} \geq \frac{2}{\sqrt{\sec\theta\csc\theta}} = \frac{2\sqrt{2}}{\sqrt{\sin 2\theta}} \geq 2\sqrt{2}$$

Since a, b, c are integers, we have $\left(\frac{c}{a}+\frac{c}{b}\right) > 2\sqrt{2}$ which gives $\left(\frac{c}{a}+\frac{c}{b}\right)^2 > 8$.

Alternative Solution 3

By simplifying and using the AM-GM inequality,

$$\left(\frac{c}{a}+\frac{c}{b}\right)^2 = c^2\left(\frac{a+b}{ab}\right)^2 = \frac{(a62+b^2)(a+b)^2}{a^2b^2} \geq \frac{2\sqrt{a^2b^2}\left(2\sqrt{ab}\right)^2}{a^2b^2} = 8$$

with equality only if $a = b$. By using the same argument as in Solution 1, a cannot equal b and the inequality is strict.

Alternative Solution 4

$$\begin{aligned}\left(\frac{c}{a}+\frac{c}{b}\right)^2 &= \frac{c^2}{a^2}+\frac{c^2}{b^2}+\frac{2c^2}{ab} \\ &= 1+\frac{b^2}{a^2}+\frac{a^2}{b^2}+1+\frac{2\left(a^2+b^2\right)}{ab} \\ &= 2+\left(\frac{a}{b}-\frac{b}{a}\right)^2+2+\frac{2}{ab}\left((a-b)^2+2ab\right) \\ &= 4+\left(\frac{a}{b}-\frac{b}{a}\right)^2+\frac{2(a-b)^2}{ab}+4 \\ &\geq 8\end{aligned}$$

with equality only if $a = b$, which (as argued previously) cannot occur.

3. *Alternative Solution 1*

For any $t, 0 \leq 4t^2 < 1 + 4t^2$, so $0 \leq \dfrac{4t^2}{1+4t^2} < 1$. Thus x, y and z must be non-negative and less than 1.

Observe that if one of xy or z is 0, then $x = y = z = 0$.

If two of the variables are equal, say $x = y$, then the first equation becomes

$$\frac{4x^2}{1+4x^2} = x$$

This has the solution $x = 0$, which gives $x = y = z = 0$ and $x = \frac{1}{2}$ which gives x $= y = z = \frac{1}{2}$.

Finally, assume that x, y and z are non-zero and distinct. Without loss of generality we may assume that either $0 < x < y < z < 1$ or $0 < x < z < y < 1$. The two proofs are similar, so we do only the first case.

We will need the fact that $f(t) = \dfrac{4t^2}{1+4t^2}$ is increasing on the interval $(0, 1)$.

To prove this, if $0 < s < t < 1$ then

$$\begin{aligned} f(t) - f(s) &= \frac{4t^2}{1+4t^2} - \frac{4s^2}{1+4s^2} \\ &= \frac{4t^2 - 4s^2}{(1+4x^2)(1+4s^2)} \\ &> 0 \end{aligned}$$

So $0 < x < y < z \;\Rightarrow\; f(x) = y < f(y) = z < f(z) = x$, a contradiction. Hence $x = y = z = 0$ and $x = y = z = \frac{1}{2}$ are the only real solutions.

Alternative Solution 2

Notice that x, y and z are non-negative. Adding the three equations gives

$$x + y + z = \frac{4x^2}{1+4x^2} + \frac{4y^2}{1+4y^2} + \frac{4z^2}{1+4z^2}$$

This can be rearranged to give

$$\frac{x(2x-1)^2}{1+4x^2} + \frac{y(2y-1)^2}{1+4y^2} + \frac{z(2z-1)^2}{1+4z^2} = 0$$

Since each term is non-negative, each term must be 0, and hence each variable is either 0 or $\frac{1}{2}$. The original equations then show that $x = y = z = 0$ and $x = y = z = \frac{1}{2}$ are the only two solutions.

Alternative Solution 3

Notice that x, y, and z are non-negative. Multiply both sides of the inequality

$$\frac{y}{1+4y^2} \geq 0$$

by $(2y-1)^2$, and rearrange to obtain

$$y - \frac{4y^2}{1+4y^2} \geq 0$$

and hence that $y \geq z$. Similarly, $z \geq x$, and $x \geq y$. Hence, $x = y = z$ and, as in Solution 1, the two solutions follow.

Alternative Solution 4

As for solution 1, note that $x = y = z = 0$ is a solution and any other solution will have each of x, y and z positive.

The arithmetic-geometric mean inequality (or direct computation) shows that

$$\frac{1 + 4x^2}{2} \geq \sqrt{1 \cdot 4x^2} = 2x$$

and hence $x \geq \dfrac{4x^2}{1 + 4x^2} = y$, with equality if and only if $1 = 4x^2$, that is, $x = \frac{1}{2}$. Similarly, $y \geq z$ with equality if and only if $y = \frac{1}{2}$ and $z \geq x$ with equality if and only if $z = \frac{1}{2}$. Adding $x \geq y, y \geq z$ and $z \geq x$ gives $x + y + x \geq x + y + z$. Thus equality must occur in each inequality, so $x = y = z = \dfrac{1}{2}$.

4. *Alternative Solution 1*

Let the center of Γ be O, the radius r. Since $BP = BC$, let

$$\theta = \measuredangle BPC = \measuredangle BCP.$$

Quadrilateral $QABC$ is cyclic, so

$$\measuredangle BAQ = 180^\circ - \theta$$

and hence

$$\measuredangle PAQ = 120^\circ - \theta.$$

Also

$$\measuredangle APQ = 180^\circ - \measuredangle APB - \measuredangle BPC = 120^\circ - \theta$$

so $PQ = AQ$ and

$$\measuredangle AQP = 2\theta - 60^\circ.$$

Again because quadrilateral $QABC$ is cyclic,

$$\measuredangle ABC = 180^\circ - \measuredangle AQC = 240^\circ - 2\theta.$$

Triangles OAB and OCB are congruent, since $OA = OB = OC = r$ and $AB = BC$. Thus

$$\measuredangle ABO = \measuredangle CBO = \frac{1}{2}\measuredangle ABC = 120^\circ - \theta.$$

We have now shown that in triangles AQP and AOB,

$$\measuredangle PAQ = \measuredangle BAO = \measuredangle APQ = \measuredangle ABO.$$

Also $AP = AB$, so $\triangle AQP \cong \triangle AOB$. Hence $QP = OB = r$.

Alternative Solution 2

Let the center of Γ be O, the radius r. Since A, P and C lie on a circle centered at B,

$$60^\circ = \measuredangle ABP = 2\measuredangle ACP, so \measuredangle ACP = \measuredangle ACQ = 30^\circ.$$

Since Q, A, and C lie on Γ,

$$\measuredangle QOA = 2\measuredangle QCA = 60^\circ.$$

So $QA = r$ since if a chord of a circle subtends an angle of 60° at the center, its length is the radius of the circle.
Now $BP = BC$, so

$$\measuredangle BPC = \measuredangle BCP = \measuredangle ACB + 30^\circ.$$

Thus

$$\measuredangle APQ = 180^\circ - \measuredangle APB - \measuredangle BPC = 90^\circ - \measuredangle ACB.$$

Since Q, A, B and C lie on Γ and $AB = BC$,

$$\measuredangle AQP = \measuredangle AQC = \measuredangle AQB + \measuredangle BQC = 2\measuredangle ACB.$$

Finally,

$$\measuredangle QAP = 180 - \measuredangle AQP - \measuredangle APQ = 90 - \measuredangle ACB.$$

So

$$\measuredangle PAQ = \measuredangle APQ$$

hence

$$PQ = AQ = r.$$

5. Rearranging the equation we get $4x^2 + 51 = 40[x]$. It is known that $x \geq [x] > x - 1$, so

$$\begin{aligned} 4x^2 + 51 = 40[x] &> 40(x-1) \\ 4x^2 - 40x + 91 &> 0 \\ (2x-13)(2x-7) &> 0 \end{aligned}$$

Hence $x < 13/2$ or $x < 7/2$. Also

$$\begin{aligned} 4x^2 + 51 = 40[x] &\leq 40x \\ 4x^2 - 40x + 51 &\leq 0 \\ (2x-17)(2x-3) &\leq 0 \end{aligned}$$

Hence $3/2 \leq x \leq 17/2$. Combining these inequalities give $3/2 \leq x < 7/2$ or $13/2 < x \leq 17/2$.

CASE 1 $3/2 \leq x < 7/2$

For this case, the possible values for $[x]$ are $1, 2$ and 3.

If $[x] = 1$ then $4x^2 + 51 = 40 \cdot 1$ so $4x^2 = -11$, which has no real solutions.

If $[x] = 2$ then $4x^2 + 51 = 40 \cdot 2$ so $4x^2 = 29$ and $x = \frac{\sqrt{29}}{2}$. Notice that $\dfrac{\sqrt{16}}{2} < \dfrac{\sqrt{29}}{2} < \dfrac{\sqrt{36}}{2}$ so $2 < x < 3$ and $[x] = 2$.

If $[x] = 3$ then $4x^2 + 51 = 40 \cdot 3$ and $x = \frac{\sqrt{69}}{2}$. But $\dfrac{\sqrt{69}}{2} > \dfrac{\sqrt{64}}{2} = 4$. So, this solution is rejected.

CASE 2 $13/2 < x \leq 17/2$

For this case, the possible values for $[x]$ are $6, 7$ and 8.

If $[x] = 6$ then $4x^2 + 51 = 40 \cdot 6$ so $x = \frac{\sqrt{189}}{2}$. Notice that $\dfrac{\sqrt{144}}{2} < \dfrac{\sqrt{189}}{2} < \dfrac{\sqrt{196}}{2}$ so $6 < x < 7$ and $[x] = 6$.

If $[x] = 7$ then $4x^2 + 51 = 40 \cdot 7$ so $x = \frac{\sqrt{229}}{2}$. Notice that $\dfrac{\sqrt{196}}{2} < \dfrac{\sqrt{229}}{2} < \dfrac{\sqrt{256}}{2}$ so $7 < x < 8$ and $[x] = 7$.

If $[x] = 8$ then $4x^2 + 51 = 40 \cdot 8$ so $x = \frac{\sqrt{269}}{2}$. Notice that $\dfrac{\sqrt{256}}{2} < \dfrac{\sqrt{269}}{2} < \dfrac{\sqrt{324}}{2}$ so $8 < x < 9$ and $[x] = 8$.

The solutions are

$$x = \frac{\sqrt{29}}{2}, \frac{\sqrt{189}}{2}, \frac{\sqrt{229}}{2}, \frac{\sqrt{269}}{2}$$

6.

$$\begin{aligned}
\sum_{n=1}^{1994}(-1)^n\frac{n^2+n+1}{n!} &= \sum_{n=1}^{1994}(-1)^n\frac{n^2}{n!}+\sum_{n=1}^{1994}(-1)^n\frac{n+1}{n!}\\
&= \sum_{n=1}^{1994}(-1)^n\frac{n}{(n-1)!}+\sum_{n=1}^{1994}(-1)^n\frac{n+1}{n!}\\
&= \sum_{n=0}^{1993}(-1)^{n+1}\frac{n+1}{n!}+\sum_{n=1}^{1994}(-1)^n\frac{n+1}{n!}\\
&= -1+\sum_{n=1}^{1993}(-1)^{n+1}\frac{n+1}{n!}+\sum_{n=1}^{1994}(-1)^n\frac{n+1}{n!}\\
&= -1+\frac{1995}{1994!}
\end{aligned}$$

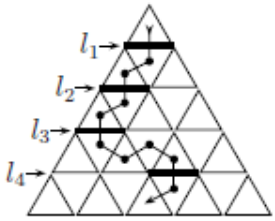
By telescoping sums.

7. We shall show that $f(n) = (n1)!$.

Label the horizontal line segments in the triangle $l_1, l_2, \ldots$ as in the diagram below. Since the path goes from the top triangle to a triangle in the bottom row and never travels up, the path must cross each of $l_1, l_2, \ldots, l_{n-1}$ exactly once. The diagonal lines in the triangle divide l_k into k unit line segments and the path must cross exactly one of these k segments for each k. (In the diagram below, these line segments have been highlighted.) The path is completely determined by the set of $n-1$ line segments which are crossed. So as the path moves from the kth row to the $(k+1)$st row, there are k possible line segments where the path could cross l_k. Since there are

$$1 \cdot 2 \cdot 3 \cdots (n-1) = (n-1)!$$

ways that the path could cross the $n-1$ horizontal lines, and each one corresponds to a unique path, we get $f(n) = (n-1)!$. Therefore $f(2005) = (2004)!$.



8. For the sake of contradiction, we assume that there is a seating arrangement such that there is no one sitting in between girls. We call a block, any group of girls (boys) sitting next to each other and sandwiched by boys(girls) from both sides.

   By assumption, each girl block has at most two girls and there is at least 2 boys in the gap between two consecutive girl blocks.

   Hence there are at least $\frac{25}{2} = 13$ girl blocks and at least 2×13 boys sitting in between the 13 gaps between girl blocks. But we only have 25 boys, a contradiction. Therefore, our assumption was wrong and it is always possible to find someone sitting between two girls.

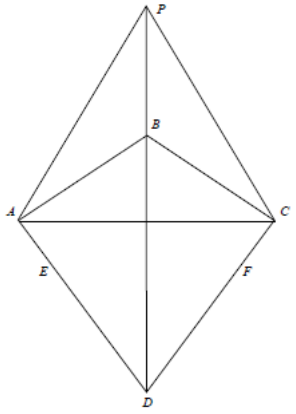
   Advanced learners can also apply the Pigeonhole principle.

9. *Alternative Solution 1*

Extend DB to a point P on the circle through A and C centered at B. Then

$$\angle CPD = \frac{1}{2}\angle CBD = \angle ADB \text{ and } \angle APD = \frac{1}{2}\angle ABD = \angle CDB,$$

so $APCD$ is a parallelogram. Now PD bisects AC so BD is an angle bisector of isosceles triangle ABC.



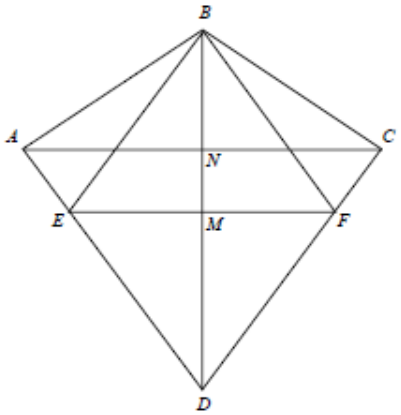
We have

$$\angle ADB = \frac{1}{2}\angle CBD = \frac{1}{2}\angle ABD = \angle CDB$$

so DB is the angle bisector of $\angle ADC$. As DB bisects the base of triangle ADC, this triangle must be isosceles and $AD = CD$.

Alternative Solution 2

Let the bisector of $\angle ABD$ meet AD at E. Let the bisector of $\angle CBD$ meet CD at F. Then $\angle FBD = \angle BDE$ and $\angle EBD = \angle BDF$, which imply $BE \parallel FD$ and $BF \parallel ED$.



Thus $BEDF$ is a parallelogram whence

$$BD \text{ intersects } EF \text{ at its midpoint } M. \tag{1.1}$$

On the other hand since BE is an angle bisector, we have $\frac{AB}{BD} = \frac{AE}{ED}$. Similarly we have $\frac{CB}{BD} = \frac{CF}{FD}$. By assumption $AB = CB$ so $\frac{AE}{ED} = \frac{CF}{FD}$ which implies $EF \parallel AC$. Thus $\triangle DEF$ and $\triangle DAC$ are similar, which implies by (1.11) that BD intersects AC at its midpoint N.

Since $\triangle ABC$ is isosceles, this implies $AC \perp BD$. Thus $\triangle NAD$ and $\triangle NCD$ are right triangles with equal legs and hence are congruent. Thus $AD = CD$.

10. If $f(x) = x^3 - x - 1 = (x-\alpha)(x-\beta)(x-\gamma)$ has roots α, β, γ standard results about roots of polynomials give

$$\begin{aligned} \alpha + \beta + \gamma &= 0, \\ \alpha\beta + \alpha\gamma + \beta\gamma &= -1, \text{ and} \\ \alpha\beta\gamma &= 1. \end{aligned}$$

Then

$$S = \frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} = \frac{N}{(1-\alpha)(1-\beta)(1-\gamma)}$$

where the numerator simplifies to

$$\begin{aligned} N &= 3 - (\alpha+\beta+\gamma) - (\alpha\beta+\alpha\gamma+\beta\gamma) + 3\alpha\beta\gamma \\ &= 3 - (0) - (-1) + 3(1) \\ &= 7. \end{aligned}$$

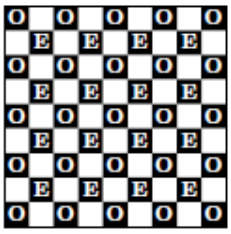
The denominator is $f(1) = -1$ so the required sum is -7.

SECTION B : 10 marks each

11. *Alternative Solution 1*

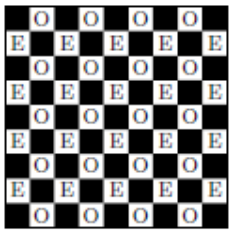
We will first count the number of ways of placing 8 mutually non-attacking rooks on black squares and then count the number of ways of placing them on white squares. Suppose that the rows of the board have been numbered 1 to 9 from top to bottom.

First notice that a rook placed on a black square in an odd numbered row cannot attack a rook on a black square in an even numbered row. This effectively partitions the black squares into a 5×5 board and a 4×4 board (squares labelled O and E respectively, in the diagram at right) and rooks can be placed independently on these two boards. There are 5! ways to place 5 non-attacking rooks on the squares labelled O and 4! ways to place 4 non-attacking rooks on the squares labelled E.



This gives 5!4! ways to place 9 mutually non-attacking rooks on black squares and removing any one of these 9 rooks gives one of the desired configurations. Thus there are $9 \cdot 5!4!$ ways to place 8 mutually non-attacking rooks on black squares.

Using very similar reasoning we can partition the white squares as shown in the diagram at right. The white squares are partitioned into two 5×4 boards such that no rook on a square marked O can attack a rook on a square mark E. At most 4 non-attacking rooks can be placed on a 5×4 board and they can be placed in $5 \cdot 4 \cdot 3 \cdot 2 = 5!$ ways. Thus there are $(5!)2$ ways to place 8 mutually non-attacking rooks on white squares. In total there are



$$9 \cdot 5!4! + (5!)2 = (9 + 5)5!4! = 14 \cdot 5!4! = 40320$$

ways to place 8 mutually non-attacking rooks on squares of the same colour.

Alternative Solution 2

Consider rooks on black squares first. We have 8 rooks and 9 rows, so exactly one row will be without rooks. There are two cases: either the empty row has 5 black squares or it has 4 black squares. By permutation these rows can be made either last or second last. In each case well count the possible number of ways of placing the rooks on the board as we proceed row by row.

In the first case we have 5 choices for the empty row, then we can place a rook on any of the black squares in row 1 (5 possibilities) and any of the black squares in row 2 (4 possibilities). When we

attempt to place a rook in row 3, we must avoid the column containing the rook that was placed in row 1, so we have 4 possibilities. Using similar reasoning, we can place the rook on any of 3 possible black squares in row 4, etc. The total number of possibilities for the first case is

$$5 \cdot 5 \cdot 4 \cdot 4 \cdot 3 \cdot 3 \cdot 2 \cdot 2 \cdot 1 = (5!)2.$$

In the second case, we have 4 choices for the empty row (but assume its the second last row). We now place rooks as before and using similar logic, we get that the total number of possibilities for the second case is

$$4 \cdot 5 \cdot 4 \cdot 4 \cdot 3 \cdot 3 \cdot 2 \cdot 1 \cdot 1 = 4(5!4!).$$

Now, do the same for the white squares. If a row with 4 white squares is empty (5 ways to choose it), then the total number of possibilities is (5!)2. Its impossible to have a row with 5 white squares empty, so the total number of ways to place rooks is

$$(5!)2 + 4(5!4!) + (5!)2 = (5 + 4 + 5)5!4! = 14(5!4!).$$

12. *Alternative Solution 1*

A prime indicates where a tangent meets AB and a double prime where it meets AC. It is given that

$$DD' = DD'', EE' = EE'' \text{ and } FF' = FF''.$$

It is required to show that arc EF is a third of the circumference as is arc DBF.

AF is the median to the hypotenuse of right triangle $AF'F''$, so that $FF' = FA$ and therefore

$$\text{arc}\, AF = 2\angle F''FA = 2(\angle FF'A + \angle FAF') = 4\angle FAF' = 4\angle FAB = 2\,\text{arc}\, BF$$

whence $\text{arc}\, FA = (2/3)\,\text{arc}\, BFA$. Similarly, $\text{arc}\, AE = (2/3)\,\text{arc}\, AEC$. Therefore, $\text{arc}\, FE$ is $2/3$ of the semicircle, or $1/3$ of the circumference as desired.

As for $\text{arc}\, DBF$,

$$\text{arc}\, BD = 2\angle BAD = \angle BAD + \angle BD'D = \angle ADD'' = (1/2)\,\text{arc}\, ACD.$$

But, $\text{arc}\, BF = (1/2)\,\text{arc}\, AF$, so $\text{arc}\, DBF = (1/2)\,\text{arc}\, FAED$. That is, $\text{arc}\, DBF$ is $1/3$ the circumference and the proof is complete.

Alternative Solution 2

Since $AE'E''$ is a right triangle, $AE = EE' = EE''$ so that $\angle CAE = \angle CE''E$. Also $AD = D'D = DD''$, so that

$$\angle CDD'' = \angle CAD = \angle CD''D.$$

As $EADC$ is a concyclic quadrilateral,

$$\begin{aligned}
180^\circ &= \angle EAD + \angle ECD \\
&= \angle DAC + \angle CAE + \angle ECA + \angle ACD \\
&= \angle DAC + \angle CAE + \angle CEE'' + \angle CE''E + \angle CDD'' + \angle CD''D \\
&= \angle DAC + \angle CAE + \angle CAE + \angle CAE + \angle CAD + \angle CAD \\
&= 3(\angle DAC + \angle DAE) \\
&= 3(\angle DAE)
\end{aligned}$$

Hence $\angle DFE = \angle DAE = 60^\circ$. Similarly, $\angle DEF = 60^\circ$. It follows that triangle DEF is equilateral.

13. *Alternative Solution 1*

We claim that f is a constant function. Suppose, for a contradiction, that there exist x and y with $f(x) < f(y)$, choose x, y such that $f(y) - f(x) > 0$ is minimal. Then

$$f(x) = \frac{xf(x) + yf(x)}{x+y} < \frac{xf(y) + yf(x)}{x+y} < \frac{xf(y) + yf(y)}{x+y} = f(y)$$

so for $f(x) < f(x^2+y^2) < f(y)$ and $0 < f(x^2+y^2) < f(y) - f(x)$, contradicting the choice of x and y. Thus, f is a constant function. Since $f(0)$ is in $\mathbb{N}$, the constant must be from $\mathbb{N}$.

Also, for any c in $\mathbb{N}$, $xc + yc = (x+y)c$ for all x and y, so $f(x) = c, c \in \mathbb{N}$ are the solutions to the equation.

Alternative Solution 2

We claim f is a constant function. Define $g(x) = f(x) - f(0)$. Then

$$g(0) = 0, g(x) \geq f(0)$$

and

$$xg(y) + yg(x) = (x+y)g\left(x^2+y^2\right)$$

for all $x, y \in \mathbb{N}$.

Letting $y = 0$ shows $g\left(x^2\right) = 0$ (in particular, $g(1) = g(4) = 0$), and letting $x = y = 1$ shows $g(2) = 0$. Also, if x, y and z in $\mathbb{N}$ satisfy $x^2 + y^2 = z^2$, then

$$g(y) \quad = \quad -\frac{y}{x}g(x) \tag{1.2}$$

Letting $x = 4$ and $y = 3$, (1.2) shows that $g(3) = 0$.

For any even number $x = 2n > 4$, let $y = n^2 - 1$. Then $y > x$ and $x^2 + y^2 = \left(n^2+1\right)^2$. For any odd number $x = 2n+1 > 3$, let $y = 2(n+1)^n$. Then $y > x$ and $x^2 + y^2 = \left((n+1)^2 + n^2\right)^2$. Thus for every $x > 4$ there is $y > x$ such that (1.2) is satisfied.

Suppose for a contradiction, that there is $x > 4$ with $g(x) > 0$. Then we can construct a sequence $x = x_0 < x_1 < x_2 < \ldots$ where $g\left(x_{i+1}\right) = -\frac{x_{i+1}}{x_i} g\left(x_i\right)$. It follows that $\left|g\left(x_{i+1}\right)\right| > \left|g\left(x_i\right)\right|$ and the signs of $g(x_i)$ alternate. Since $g(x)$ is always an integer, $\left|g\left(x_{i+1}\right)\right| \geq \left|g\left(x_i\right)\right| + 1$. Thus for some sufficiently large value of i, $g(x_i) < -f(0)$, a contradiction.

As for Proof 1, we now conclude that the functions that satisfy the given functional equation are $f(x) = c, c \in N$.

Alternative Solution 3

Suppose that W is the set of nonnegative integers and that $f : W \to W$ satisfies:

$$xf(y) + yf(x) \quad = \quad (x+y)f\left(x^2+y^2\right)$$

We show that f is a constant function.

14. *Alternative Solution 1*

Let $f(x,y,z) = (x^3-x)+(y^3-y)+(z^3-z)$. The first equation above is equivalent to $f(x,y,z) = 0$. If $x, y, z \geq 1$, then $f(x,y,z) \geq 0$ with equality only if $x = y = z = 1$.

But if $x = y = z = 1$, then the second equation is not satisfied. So in any solution to the system of equations, at least one of the variables is less than 1. Without loss of generality, suppose that $x < 1$. Then

$$x^2 + y^2 + z^2 > y^2 + z^2 \geq 2yz > yz > xyz.$$

Therefore the system has no real positive solutions.

Alternative Solution 2

We will show that the system has no real positive solution. Assume otherwise.

The second equation can be written $x^2 - (yz)x + (y^2 + z^2)$. Since this quadratic in x has a real solution by hypothesis, its discrimant is nonnegative. Hence

$$y^2z^2 - 4y^2 - 4z^2 \geq 0.$$

Dividing through by $4y^2z^2$ yields

$$\frac{1}{4} \geq \frac{1}{y^2} + \frac{1}{z^2} \geq \frac{1}{y^2}.$$

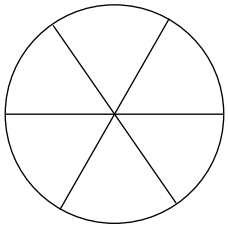
Hence $y^2 \geq 4$ and so $y \geq 2$, y being positive. A similar argument yields $x, y, z \geq 2$. But the first equation can be written as

$$x(x^2 - 1) + y(y^2 - 1) + z(z^2 - 1) = 0,$$

contradicting $x, y, z \geq 2$. Hence, a real positive solution cannot exist.

15. (a) Let n be the number of people at the party. Not counting themselves, a person can have $0, 1, \cdots, n-2$ or $n-1$ friends. If all n people had a different number of friends, then one person would have 0 friends and another would have $n-1$. This is not possible since the person with $n-1$ friends is friends with everyone at the party, including the person with 0 friends.

(b) To prove that the final statement is always true, we first divide the circle into six equal sectors as shown:



Allowing each sector to be a pigeonhole and each dart to be a pigeon, we have seven pigeons to go into six pigeonholes. By pigeonhole principle, there is at least one sector containing a minimum of two darts. Since the greatest distance between two points lying in a sector is 10 units, the statement is proven to be true in any case.

In fact, it is also possible to prove the scenario with only six darts. In such a case, the circle is this time divided into five sectors and all else follows. However, take note that this is not always true anymore with only five darts or less.